Respiration

Clinical Investigations

Respiration 2024;103:723–732
DOI: 10.1159/000540780

Received: April 29, 2024
Accepted: July 30, 2024
Published online: August 13, 2024

Deventilation Syndrome in Patients with Chronic Obstructive Pulmonary Disease Using Nocturnal Noninvasive Ventilation: What Are the Underlying Mechanisms?

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Keywords
Chronic obstructive pulmonary disease · Noninvasive ventilation · Dyspnea · Deventilation syndrome

Abstract
Introduction: Patients with chronic obstructive pulmonary disease (COPD) commonly experience severe dyspnea after discontinuation of nocturnal noninvasive ventilation (NIV), known as deventilation syndrome (DVS), which negatively affects quality of life. Despite various hypotheses, the precise mechanisms of DVS remain unknown. ***Methods:*** An observational pilot study was performed monitoring 16 stable COPD patients before, during, and after an afternoon nap on NIV. Seven patients experienced DVS (Borg Dyspnea Scale ≥5), while nine served as controls (Borg Dyspnea Scale ≤2). Hyperinflation was evaluated through inspiratory capacity (IC) measurements and end-expiratory lung impedance (EELI) via electrical impedance tomography. Respiratory muscle activity was assessed by diaphragmatic surface electromyography (sEMG). ***Results:*** Post-NIV dyspnea scores were significantly higher in the DVS group (5 [3–7] vs. 0 [0–1.5], $p < 0.001$). IC values were lower in the DVS group compared to controls, both pre-NIV (54 [41–63] vs. 88 [72–94] %pred., $p = 0.006$) and post-NIV (45 [40–59] vs. 76 [65–82] %pred., $p = 0.005$), while no intergroup difference was seen in IC changes pre- and post-NIV. EELI values after NIV indicated a tendency towards lower values in controls and higher values in DVS patients. sEMG amplitudes were higher in the DVS group within the first 5-min post-NIV (221 [112–294] vs. 100 [58–177]% of baseline, $p = 0.030$). ***Conclusion:*** This study suggests that it is unlikely that DVS originates from the inability to create diaphragmatic muscle activity after NIV. Instead, NIV-induced hyperinflation in individuals with static hyperinflation may play a significant role. Addressing hyperinflation holds promise in preventing DVS symptoms in COPD patients.

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Introduction

Chronic obstructive pulmonary disease (COPD) is a progressive respiratory disease that is characterized by airway obstruction, leading to symptoms such as dyspnea and cough [1]. Patients with COPD, particularly those in severe stages, are at risk of developing hypercapnic respiratory failure [2]. The nocturnal application of chronic noninvasive ventilation (NIV) emerges as an effective treatment for these patients, especially when aimed at achieving normocapnia (or the lowest possible carbon dioxide levels) using high inspiratory positive airway pressure (IPAP) levels, often referred to as high-intensity NIV [3]. In COPD, high-intensity chronic NIV not only improves gas exchange but also enhances patient-related outcomes, such as quality of life and survival [4].

Despite these positive outcomes, patients on chronic (high-intensity) NIV may experience acute dyspnea upon discontinuation, a phenomenon called deventilation syndrome (DVS) [5]. DVS is highly prevalent among COPD patients on chronic NIV, with its occurrence estimated to be between 28 and 58% [6–8]. DVS contributes to more dyspnea throughout the day, diminished exercise capacity, and a reduction in quality of life [6–8].

Several hypotheses have been proposed that could clarify the underlying mechanisms of DVS. One plausible explanation suggests that NIV leads to dynamic hyperinflation due to the patient's inability to fully exhale before the next inspiration occurs. This phenomenon increases residual volume and decreases inspiratory capacity (IC), which can induce dyspnea [9]. Lüthgen et al. [7] performed an explorative study in 11 DVS patients and 20 controls, and their results support this hypothesis. Their findings revealed that patients who suffered from DVS had a significantly higher mean airway resistance and an increased ratio between residual volume and total lung capacity compared to controls. Similarly, Adler et al. [6] hypothesized that DVS is the consequence of hyperinflation induced by high IPAP levels during NIV, potentially leading to patient-ventilator asynchronies (PVA) and morning dyspnea. They adjusted ventilator settings based on a polysomnography by lowering IPAP levels to prevent PVA. With this method, they improved PVA and dyspnea outcomes. Another plausible cause of DVS is the suppression of respiratory muscle activity during NIV [10], potentially leading to difficulties in reactivating the respiratory muscles after NIV due to poor muscle strength, which could contribute to DVS [11]. This hypothesis is supported by Schellenberg et al. [8] who performed an observational study including 39 DVS patients and 28 controls, and found reduced respiratory muscle strength in the DVS group compared to controls.

Altogether, several hypotheses about the underlying mechanisms of DVS exist, but the exact mechanisms of DVS remain unknown. The aim of this observational study was to gain more knowledge about the underlying mechanisms of DVS in patients with severe COPD on chronic NIV by measuring (changes in) lung volumes and diaphragmatic muscle activity and compare patients experiencing DVS symptoms to those who do not.

Methods

Study Design

This case-control pilot study was performed from September 2019 to September 2023. The target sample size was 10 patients per group. However, study inclusion was severely delayed during the COVID-19 pandemic, and ultimately, the study was closed prematurely due to logistical challenges arising from the pandemic. Patients had to meet the following criteria to be included ≥18 years of age and COPD GOLD III–IV for which chronic NIV was already started. Patient in the DVS group were required to have symptoms of DVS, as defined by a Borg Dyspnea Scale ≥5 after withdrawal of their NIV in the morning. Furthermore, DVS patients required to experience symptoms following an afternoon nap, although with the condition that the Borg Dyspnea Scale requirement was reduced to at least 3. This adjustment was made for practical reasons, as we decided to research patients following an afternoon nap, during which DVS symptoms tend to be less pronounced. Patients in the control group had no DVS symptoms (defined as Borg Dyspnea Scale ≤2). Exclusion criteria included a COPD exacerbation or a modification to the NIV within 2 weeks preceding the study, having a poor compliance with NIV (usage <4 h/day), other disorders, leading to respiratory failure and having a cardiac pacemaker or other implanted devices. The study protocol was reviewed and approved by the Medical Ethics Committee of the University Medical Center Groningen (METc 2018/290). Written informed consent was obtained from all study participants.

Study Procedures and Data Collection

Study participants were invited to the outpatient clinic or were visited by a researcher for measurements at home. To assess hyperinflation, we conducted both IC measurements using a MasterScreen Pneumo spirometer

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(Vyaire Medical Inc., IL, USA) and electrical impedance tomography (EIT) assessment using a Swisstom BB2 Device (SenTec AG (formerly Swisstom AG), Landquart, Switzerland) [12]. Respiratory muscle activity was assessed with surface electromyography (sEMG) measurements of the frontal diaphragm using two Ag/AgCl electrodes (Kendall ECG electrodes sized 30 mm × 24 mm, Cardinal Health, Dublin, OH, USA) placed bilaterally and subcostally along the midclavicular line, with a ground electrode placed on the sternum, utilizing a Dipha-16 system (DEMCON Macawi Respiratory Systems, Enschede, The Netherlands) [13]. PVA was evaluated by comparing sEMG recordings with pressure data from the ventilator, which was simultaneously obtained using a pressure transducer positioned between the NIV mask and tube (DEMCON Macawi Respiratory Systems, Enschede, The Netherlands) [14]. Transcutaneous partial pressure of carbon dioxide ($PtcCO_2$) was monitored using a V-Sign Sensor on the earlobe (SenTec AG, Therwil, Switzerland). Additionally, participants were instructed to rate their dyspnea on a 10-point Borg Dyspnea Scale. Continuous baseline measurements of EIT, sEMG, and $PtCO_2$ were performed during spontaneous breathing in both recumbent and semi-recumbent position, each for at least 5 min. Subsequently, participants were asked to sleep with their own home ventilator and settings, with continuous monitoring of EIT, sEMG, and $PtcCO_2$. Upon waking and discontinuation of NIV, participants maintained a still position for 2 min. Following this period, IC and Borg dyspnea score were determined. These assessments were repeated 15- and 30-min post-awakening and subsequently every 30 min until relief from dyspnea-related symptoms. Throughout this period, EIT, sEMG, and $PtcCO_2$ were continuously measured. Finally, the ventilator was read out to assess NIV parameters.

Analyses and Statistics

EIT, sEMG, and pressure data were imported into MATLAB (v2023b, The MathWorks Inc., Natick, MA, USA) for further analysis. For comprehensive details regarding signal processing and postprocessing, please refer to online supplementary File 1 (for all online suppl. material, see https://doi.org/10.1159/000540780).

The global impedance signal of the EIT data was used to analyze the end-expiratory lung impedance (EELI) as a measure of hyperinflation [12, 15, 16] and tidal impedance variation (TIV), representing the impedance change during a tidal breath, as a surrogate for tidal volume [12, 17]. Given the substantial dependence of EELI on both the patient's position and signal quality, the period immediately after NIV withdrawal was compared to a period before NIV initiation with the same position and signal quality. The mean EELI and TIV before NIV initiation served as the baseline. For each patient, both the EELI and TIV after NIV were measured as a percentage relative to their baseline values.

Respiratory muscle activity was assessed by analyzing diaphragmatic sEMG amplitudes, defined as the difference between maximum and minimum values during one breath. The baseline sEMG amplitude was determined by calculating the mean value over a period of at least 5 min during spontaneous breathing. The amplitudes during the 30-min period immediately following NIV withdrawal were averaged in 5-min intervals, excluding any periods with noise interference or instances when the patient executed an IC maneuver. For each patient, both during and post-NIV, sEMG amplitudes were represented as a percentage relative to the established baseline.

PVA assessment included ineffective efforts and delayed cycling, as these asynchronies are linked to hyperinflation [18, 19]. The frequency of PVAs was calculated as a percentage of all breaths analyzed. Additionally, the inspiratory time (Ti) relative to the total respiratory time of each breath, derived from the ventilator pressure, was averaged.

Mann-Whitney U tests (continuous variables) or Fisher's exact tests (binary variables) were performed to compare outcome parameters from DVS patients and controls. Values were presented as median (Q1–Q3), unless otherwise stated. A p value <0.05 was considered as statistically significant. For endpoints with small sample sizes in each group ($n \leq 3$), we conducted only descriptive analysis and did not provide p values.

Results

Sixteen patients were included in the study: seven with DVS and nine without complaints of DVS who served as controls. We did not observe any major differences between groups (Table 1). Dyspnea scores were not different between groups at baseline (controls: 1 [0–2] vs. DVS: 3 [0–3], $p = 0.299$) but were significantly higher directly after NIV in the DVS group: 5 [3–7] versus controls: 0 [0–1.5], $p < 0.001$. No difference in length of the afternoon nap was observed between groups: controls 109 [84–124] minutes versus DVS 119 [105–120] minutes, $p = 1.000$.

No significant differences in ventilator parameters were found between groups (Table 2). Additionally, the

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Table 1. Characteristics of study participants

	Controls (n = 9)	DVS patients (n = 7)	p value
Baseline			
Sex (female), n (%)	5 (55.6)	5 (71.4)	0.633
Age, years	65 [64–71]	62 [59–66]	0.174
BMI, kg/m^2	26.8 [21.7–35.0]	26.8 [22.5–32.1]	1.000
Smoking status			
Active smoker, n (%)	2 (22.2)	1 (14.3)	1.000
Pack years	38.5 [27.8–56.8]	32.0 [15.0–55.0]	0.397
Lung function			
FEV1, %pred.	21 [16–35]	30 [18–35]	0.536
FEV1/FVC, %	28 [25–31]	33 [26–39]	0.174
TLC, %pred.	132 [123–140]	132 [102–134]	0.536
RV, %pred.	184 [166–212]	237 [153–263]	0.351
RV/TLC, %	58 [54–66]	65 [55–69]	0.397
ABG before start NIV			
$PaCO_2$, kPa	7.2 [6.8–8.4]	7.0 [6.9–7.6]	0.918
Bicarbonate, mmol/L	32.0 [30.5–37.0]	35.0 [31.0–38.0]	0.681
ABG after start NIV			
$PaCO_2$, kPa	6.2 [5.9–6.9]	6.7 [5.8–7.1]	0.867
Bicarbonate, mmol/L	30.0 [27.8–32.5]	32.0 [28.0–33.0]	0.613
Specifics of noninvasive ventilation			
Duration of NIV, months	12.0 [4.8–27.3]	24.5 [12.0–51.0]	0.351
Settings			
IPAP, cm H_2O	25.0 [22.0–26.0]	23.0 [21.0–27.0]	0.837
EPAP, cm H_2O	6.0 [4.5–8.0]	6.0 [5.0–9.0]	0.758
BURR, min^{-1}	12.0 [10.5–14.0]	14.0 [10.0–14.0]	0.681
Oxygen treatment			
During daytime, L/min	1.0 [0.5–1.5]	1.0 [0.0–2.0]	0.837
During NIV, L/min	1.0 [1.0–2.5]	1.5 [1.0–2.0]	0.758

Results are presented as median (Q1–Q3) unless otherwise stated. BMI, body mass index; DVS, deventilation syndrome; FEV1, forced expiratory volume in ones; FVC, forced vital capacity; TLC, total lung capacity; RV, residual volume; ABG, arterial blood gas; NIV, noninvasive ventilation; $PaCO_2$, partial pressure of carbon dioxide; IPAP, inspiratory positive airway pressure; EPAP, expiratory positive airway pressure; BURR, backup respiratory rate.

analysis revealed no differences in transcutaneous carbon dioxide levels among the groups, whether measured at baseline, during, or after NIV (Table 3).

Measurements of Hyperinflation

Electrical Impedance Tomography

Due to unstable recordings, only a limited number of measurements were deemed suitable for EELI calculation within the first 5 min following NIV withdrawal, i.e., four patients in the control group and 3 patients within the DVS group. Following NIV, the control group exhibited a tendency toward decreased EELI values (median 96.9%, ranging from 94.8% to 100.7% of the baseline), whereas the DVS group displayed a trend toward increased values (median 101.4%, ranging from 96.0% to 104.3% of the baseline). In the 7 patients eligible for EIT analysis, similar changes in TIV values following NIV were found. The median of the control group was 102%, ranging from 80 to 128% of the baseline, while the DVS group had a median value of 113%, ranging from 104 to 124% of the baseline.

Inspiratory Capacity

Table 4 and Figure 1 show that the IC is lower, both at baseline and after NIV, in the DVS group compared to controls. Immediate post-NIV IC values were reduced compared to baseline in both the control (−11% [−14% to 1%] of predicted) and in the DVS

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Table 2. Ventilator data during the study measurements

	Controls (n = 9)	DVS patients (n = 7)	p value
Expired tidal volume, mL	769 [549–859]	600 [480–658]	0.091
Respiratory rate, min^{-1}	13 [10.5–14.5]	14 [13–14]	0.408
Minute ventilation, L/min	9.5 [8.8–11.7]	7.4 [6.9–9.6]	0.142
% patient triggered breaths	35 [8–64]	38 [25–78]	0.536
Non-intentional leak, L/min	7.7 [4.2–12.5]	5.5 [1.0–14.5]	0.689
IPAP, cm H_2O	24.1 [22.0–26.1]	23.1 [22.0–26.8]	0.918
EPAP, cm H_2O	6.0 [4.5–6.5]	6.0 [5.0–9.0]	0.606
Compliance before measurement, h/day	7.7 [7.1–10.5]	7.7 [5.4–8.6]	0.408

Results are presented as median (Q1–Q3). DVS, deventilation syndrome; IPAP, inspiratory positive airway pressure; EPAP, expiratory positive airway pressure.

Table 3. Transcutaneous carbon dioxide levels (in kPa) during the study measurements

	Controls (n = 9)	DVS patients (n = 7)	p value
Baseline	6.1 [5.3–7.1]	6.2 [5.6–7.5]	0.408
End of NIV	5.6 [5.3–6.6]	5.9 [5.2–7.3]	0.867
After afternoon sleep with NIV			
At 0 min	5.3 [5.0–6.6]	5.8 [5.2–7.0]	0.408
At 15 min	6.2 [5.6–6.6]	6.0 [5.6–7.5]	0.837
At 30 min	6.1 [5.5–6.6]	6.0 [5.6–7.8]	0.694

Results are presented as median (Q1–Q3). DVS, deventilation syndrome; NIV, noninvasive ventilation.

Table 4. Inspiratory capacity measurements (both in liters and percentage of predicted) during the study

	Controls (n = 8)	DVS patients (n = 7)	p value
Baseline			
L	1.7 [1.4–2.1]	1.1 [0.9–1.6]	0.054
% pred.	88 [72–94]	54 [41–63]	0.006
After afternoon sleep with NIV			
At 0 min			
L	1.5 [1.4–2.0]	1.1 [1.0–1.5] (n = 6)	0.059
% pred.	76 [65–82]	45 [40–59] (n = 6)	0.005
At 15 min			
L	1.4 [1.3–1.9] (n = 7)	1.1 [0.8–1.4]	0.026
% pred.	74 [61–79] (n = 7)	46 [40–54]	0.002
At 30 min			
L	1.5 [1.2–1.9] (n = 7)	1.0 [0.8–1.1]	0.007
% pred.	67 [66–82] (n = 7)	38 [34–55]	0.001

Results are presented as median (Q1–Q3). One control patient was unable to conduct inspiratory capacity measurements due to severe dyspnea resulting from the use of the nose clip. DVS, deventilation syndrome; NIV, noninvasive ventilation.

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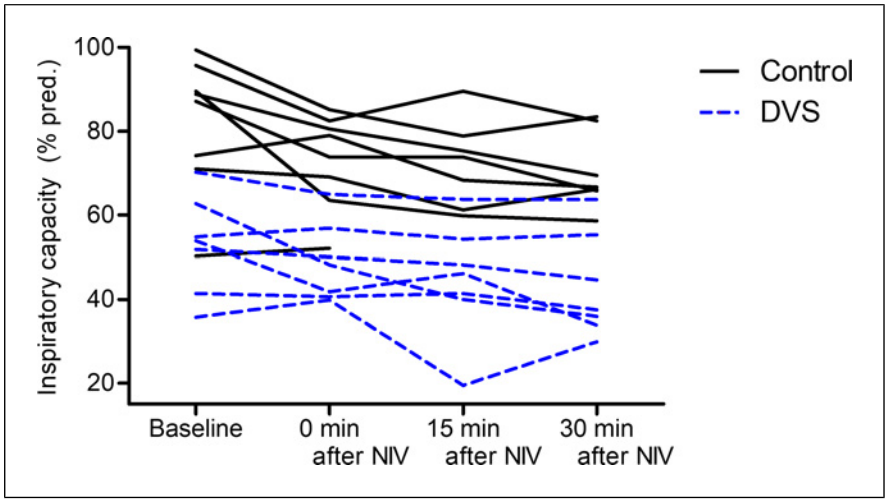

Fig. 1. Individual levels of IC (in percentage of predicted) during the study measurements, specified per patient group. DVS, deventilation syndrome; NIV, noninvasive ventilation.

group (−3% [−13% to 1%] of predicted), but this decrease did not differ significantly between groups (p = 0.491).

Respiratory Muscle Activity

The quality of the sEMG data was sufficient for data analysis in 5 controls and 6 DVS patients. For the other patients, data quality was deemed insufficient due to significant challenges in distinguishing between breathing effort and noise during visual inspection of the sEMG data. Table 5 illustrates that sEMG amplitudes decrease during NIV compared to baseline in both groups. Following NIV withdrawal, sEMG amplitudes are significantly higher in the first 5 min in the DVS group compared to controls (Table 5; Fig. 2). Figure 3 illustrates a representative sEMG tracing from a patient in the control and another in the DVS group, clearly highlighting the increased amplitude observed in the DVS patient following NIV compared to baseline.

Patient-Ventilator Asynchronies

Out of the 5 controls and 6 DVS patients who provided sEMG data of satisfactory quality, 3 individuals from each group were eligible for PVA analysis. For the other patients, challenges arose in accurately discerning between breathing effort and noise in the sEMG data due to the low signal-to-noise ratio during NIV (n = 2) or technical issues prevented the measurement of sEMG and/or pressure data (n = 3). A trend toward more ineffective efforts was seen in DVS patients (median 2.2%, ranging from 1.2 to 6.0%) compared to controls (median 0%, ranging from 0 to 1.6%). Similar outcomes were observed between the groups in delayed cycling (DVS: median of 1.8%, ranging from 0.2 to 3.4% vs. control: median of 1.4%, ranging from 0.0 to 4.8%) and the Ti/total respiratory time ratio (DVS: median of 0.23, ranging from 0.22 to 0.24 versus control: median of 0.21, ranging from 0.19 to 0.30.

Discussion

The aim of this study was to gain more insight into the underlying mechanisms of DVS in COPD patients on chronic NIV. The results revealed noteworthy differences, particularly in post-NIV sEMG amplitudes, where DVS patients exhibited higher values compared to controls. These findings may challenge the notion that DVS originates from the incapability to create diaphragmatic muscle activity after NIV. Additionally, IC measurements showed consistently lower values in the DVS group, both at baseline and post-NIV. This indicates that static hyperinflation, combined with NIV-induced hyperinflation, plays a pivotal role in the manifestation of DVS in COPD patients.

Our results, building upon previous research [10], confirmed the reduction in respiratory muscle activity during NIV. This phenomenon led to the speculation that DVS patients might encounter challenges in reactivating their diaphragm after NIV. We are the first study to monitor sEMG in these patients, revealing no difficulties in diaphragm reactivation after NIV. In fact, higher diaphragm sEMG amplitudes were found after NIV in DVS patients compared to baseline, while control patients maintained similar sEMG amplitudes, thereby questioning the aforementioned hypothesis. It is important to note that sEMG monitors respiratory

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Table 5. sEMG amplitudes (in percentage relative to baseline) during NIV and the first 30 min following NIV withdrawal, averaged per 5-min interval

sEMG amplitude	Controls (n = 5)	DVS patients (n = 6)	p value
During NIV, % baseline	4 [0–74] (n = 3)	26 [10–36] (n = 4)	0.629
0–5 min after NIV, % baseline	100 [58–177]	221 [112–294]	0.030
5–10 min after NIV, % baseline	119 [109–149] (n = 3)	171 [123–258]	0.167
10–15 min after NIV, % baseline	82 [65–189]	153 [132–215]	0.177
15–20 min after NIV, % baseline	130 [70–244]	200 [136–303]	0.126
20–25 min after NIV, % baseline	100 [79–219] (n = 3)	147 [102–257]	0.714
25–30 min after NIV, % baseline	119 [78–211] (n = 4)	144 [92–283]	0.476

Values represented as median (min–max). sEMG amplitudes during NIV could not be calculated in 4 patients due to insufficient quality (n = 2) or technical difficulties (n = 2). For some intervals after NIV withdrawal, no amplitudes could be calculated due to noisy interference or IC maneuvers. DVS, deventilation syndrome; NIV, noninvasive ventilation; sEMG, surface electromyography.

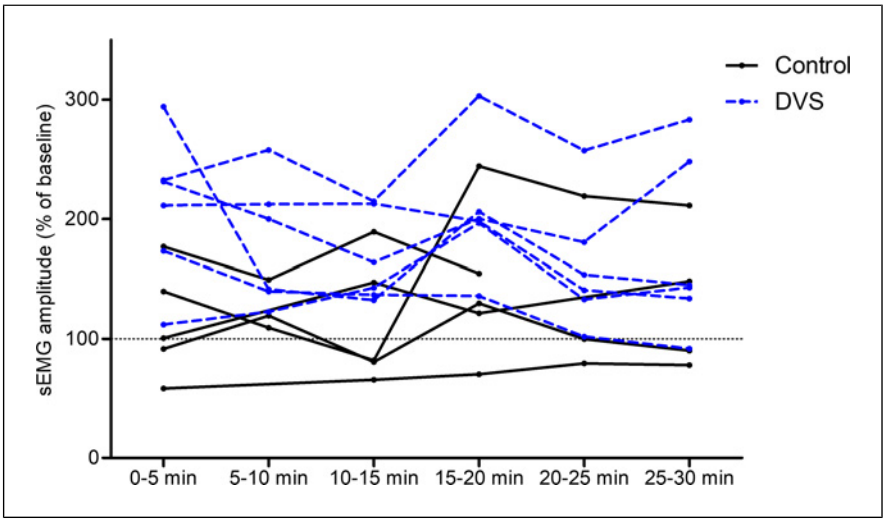

Fig. 2. Individual levels of sEMG amplitudes (in percentage of baseline) during the first 30 min after NIV withdrawal, averaged over 5-min periods, specified per patient group. The dotted black line represents the sEMG amplitude at baseline. DVS, deventilation syndrome; sEMG, surface electromyography.

neural drive [13] and does not account for respiratory mechanical output, such as transdiaphragmatic pressure, diaphragm excursion, or tidal volumes. The efficiency or the neuromechanical coupling of the diaphragm in DVS patients could theoretically be altered [20], potentially due to hyperinflation, which could reduce diaphragm strength [21]. This notion is suggested by Schellenberg et al. [8], who at baseline, observed reduced respiratory muscle strength in DVS patients compared to controls through maximal inspiratory mouth pressure measurements. Notably, they did repeat this measurement after NIV. If the force generating capacity of the diaphragm, and consequently its efficiency, would be reduced in DVS patients due to hyperinflation, it would require increased electrical activity for a specific respiratory output. However, we did not directly measure output. Our data suggest that DVS does not originate from insufficient neural respiratory drive. However, we recommend to conduct further research that simultaneous monitors both sEMG and a direct measure of respiratory output to comprehensively evaluate diaphragm efficiency and neuromechanical coupling in DVS patients.

Hyperinflation has been suggested as an underlying cause of DVS in previous studies, although direct measurement was lacking [6, 8, 22]. Lüthgen et al. [7] also proposed hyperinflation as a cause of DVS based on elevated airway resistance and residual volume and

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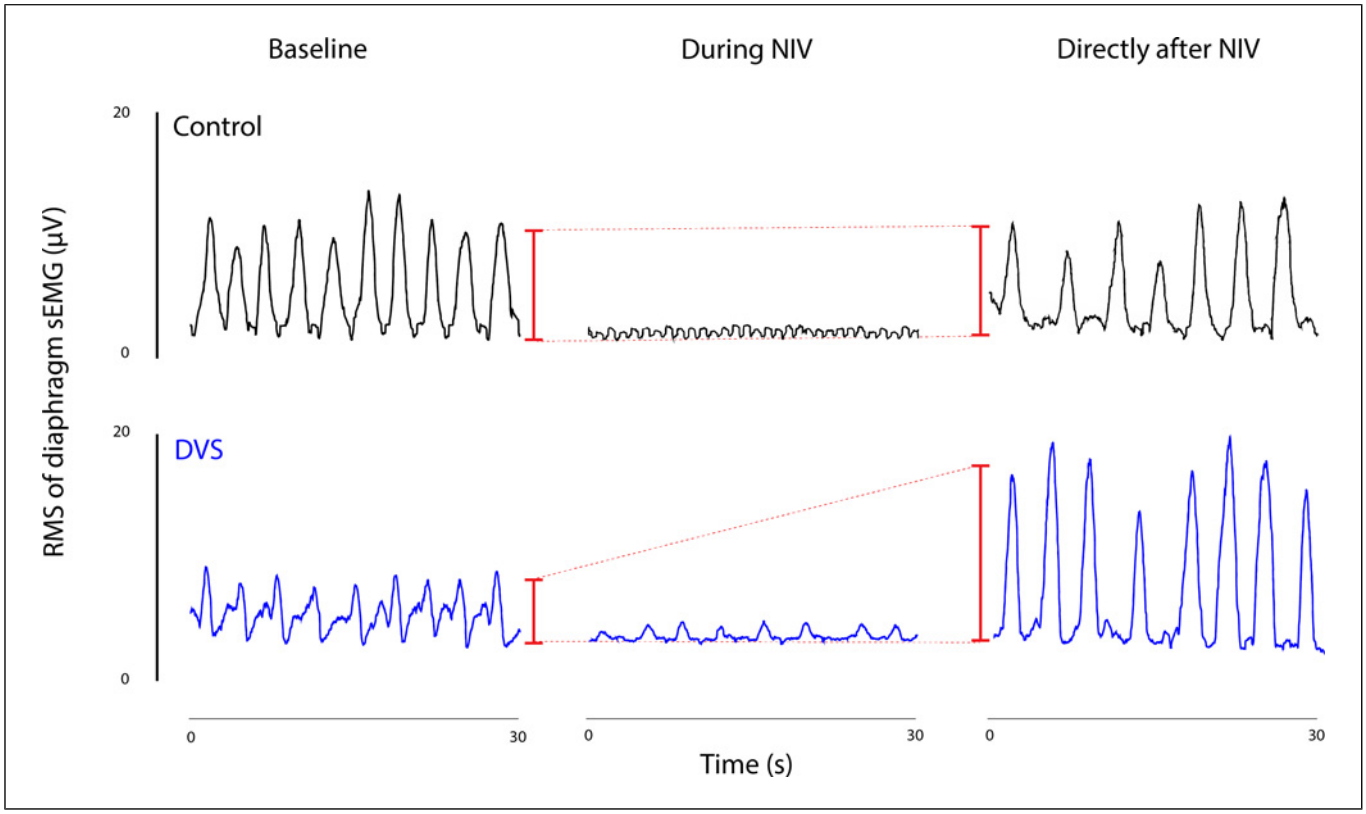

Fig. 3. Representative tracing of the preprocessed diaphragm sEMG signal for a patient in the control group and a patient with DVS. The red lines represent the sEMG amplitudes, both at baseline and directly after NIV. DVS, deventilation syndrome; NIV, noninvasive ventilation; RMS, root mean square; sEMG, surface electromyography.

total lung capacity values in DVS patients. Additionally, their study stands out as the sole investigation to conduct spirometry pre- and post-NIV. They found lower vital capacity in DVS patients at baseline, aligning with our results indicating lower IC at baseline and persisting post-baseline, with no intergroup differences pre- and post-NIV. Our IC measurements reveal a form of NIV-induced dynamic hyperinflation, as evidenced by lower post-NIV values compared to baseline. This change does not exhibit significant differences between the groups. There appears to be a trend towards more hyperinflation after NIV in the DVS group, as suggested by the EIT data, showing higher EELI values in the DVS group. Additionally, there is a trend towards increased ineffective efforts in the DVS group in the PVA analysis. This may suggest the involvement of hyperinflation as well, since increased dynamic hyperinflation and intrinsic positive end-expiratory pressure raise the breathing effort needed to trigger the ventilator. The potential explanation for dyspnea in DVS patients lies in their lower baseline IC, indicating static hyperinflation. The further reduction of IC induced by NIV could serve as a turning point in these patients, and could potentially bring the level of hyperinflation close to the dyspnea threshold, representing an inflection point beyond which dyspnea levels become intolerable [23, 24].

Strategies for preventing DVS should likely prioritize the reduction of both static and NIV-induced dynamic hyperinflation. Various general treatment options to address static hyperinflation are available, for example, optimal pharmacological treatment with bronchodilators [1]. Additionally, bronchoscopic lung volume reduction treatment emerges as an effective treatment for a very specific group of emphysematous COPD patients [25, 26]. In regard to reducing NIV-induced dynamic hyperinflation, adjustment of ventilator settings is a good starting point: this involves reduction of the backup respiratory rate and/or the Ti to offer the patient sufficient time to exhale.

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Furthermore, lower IPAP settings leading to smaller tidal volumes and reduced PVA could reduce dynamic hyperinflation [6]. In addition, specific NIV modes in this patient group may hold promise, such as pursed lip breathing ventilation, which prevents airway collapse during expiration and has shown to reduce dyspnea or exercise tolerance [7, 22]. Alternatively, ventilators employing a forced oscillation technique to measure expiratory flow limitation and adjust EPAP accordingly could be of interest [27, 28]. At last, easy-to-use and low-risk approaches, not to prevent DVS but alleviate its symptoms after NIV, involve the use of a positive expiratory pressure device or a hand-held fan [29, 30].

This study has several limitations, with the foremost being the small number of patients. Despite DVS being a prevalent symptom in clinical practice, the complexity of inclusion criteria (i.e., requiring DVS complaints following an afternoon nap) and the advanced disease severity in these patients posed challenges, impeding their ability and willingness to participate. Nevertheless, despite the limited patient cohort, we have identified noteworthy differences between the groups. While EIT is a promising monitoring tool in the intensive care unit [31], its application in spontaneously breathing and awake patients has proven to be challenging, resulting in a limited number of EIT datasets suitable for analysis. Furthermore, sEMG data analysis faced difficulties, particularly during NIV, as reduced sEMG activity led to low signal-to-noise ratios [10, 32]. Consequently, emphasis should be placed on individual trends during and after NIV rather than specific numerical values. Furthermore, cross-talk of other muscles, such as the abdominal muscles, could not be entirely ruled out. Additionally, our measurements were conducted following an afternoon nap, potentially leading to an underestimation of our results given that DVS patients typically manifest more symptoms after a full night's sleep. Lastly, it is worth noting that not all patients had recent lung function measurements. More up-to-date values could have contributed to clearer representation of static hyperinflation in the DVS group, particularly regarding body plethysmographic values.

Conclusion

This observational pilot study suggests that DVS does not originate from the inability to create diaphragmatic muscle activity after NIV. Rather, hyperinflation appears to be an important underlying mechanism in DVS. Addressing hyperinflation could potentially prevent DVS symptoms in these patients.

Statement of Ethics

The study protocol was reviewed and approved by the Medical Ethics Committee of the University Medical Center Groningen, Approval No. (METc 2018/290). Written informed consent was obtained from all study participants. The trial was registered on clinicaltrials.gov (NCT03503123).

Conflict of Interest Statement

J. Elshof reports grants and speaking fees from Vivisol B.V. and grants from Fisher & Paykel, outside the submitted work. E. Oppersma reports grants from Vivisol B.V., Löwenstein B.V., and Sencure B.V., outside the submitted work. A. Torres reports grants from the government of Catalonia (CERCA program) and the Spanish Ministry of Science and Innovation, outside the submitted work. P.J. Wijkstra reports consulting fees from Philips B.V., grants from ResMed Ltd., and a leadership role (treasurer) in the ERS board, outside the submitted work. M.L. Duiverman reports grants from Philips B.V., Fisher & Paykel, Vivisol B.V., ResMed Ltd., Löwenstein B.V., and Sencure B.V., speaking fees from Vivisol B.V., ResMed Ltd., Novartis, Chiesi, Breas, and AstraZeneca, and a leadership role (chair group NIV) in ERS assembly 2. The other authors declare that they have no competing interests.

Funding Sources

The authors have not received any financial support.

Author Contributions

J.E.: methodology, investigation, software, formal analysis, writing – original draft, and writing – review and editing. E.O. and J.J.W.: methodology, software, and writing – review and editing. G.B.: investigation, writing – review and editing. P.M.M.: methodology, investigation, writing – review and editing. A.T.: methodology, software, and writing – review and editing. P.J.W.: methodology, supervision, and writing – review and editing. M.L.D.: conceptualization, methodology, supervision, and writing – review and editing. All authors read and approved the final manuscript.

Data Availability Statement

The data that support the findings of this study are not publicly available due to their containing information that could compromise the privacy of research participants but are available from J. Elshof upon reasonable request.

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